Genuine Multipartite Entanglement Analysis — Molecular System

MOLECULE

Ethylene (C2H4)
Basis set   : cc-pVDZ
Charge   : 0
Multiplicity : 1
Total e-   : 16
n-body order : 2

GROUND STATE ENERGIES

E0 (Ha) : -78.05755506
E1 (Ha) : -77.89240700
Gap (Ha) : 0.16514806
HF (Ha) : -78.03837969

CONVEX-ROOF METHODS

Methods run   : gcg
Restarts   : 8

C_LB : coherence witness bound
C_expo: unitary opt (exp map)
C_stf : Stiefel manifold (Alg. B)
C_UB : eigendecomposition bound

ACTIVE SPACE CAS(8,5)

Frozen core orbs : 4
Frozen e-   : 8
Active e-   : 8
Active orbs   : 5
Spin-orbitals   : 10
Hilbert dim   : 25

SPIN PROPERTIES

<S2>   : 0.000000
S   : 0.0000
<Sz>   : 0.000000
Multiplicity : 1
Spin state   : Singlet (S=0)

GME CRITERIA

C_GME = sqrt(2 * min_A (1 - Tr(rho_A^2)))

Convex roof (mixed):
  C_GME(rho) = min Sum p_i C_GME(|psi_i>)

Bounds ordering:
  C_LB <= C_GME <= C_UB

DOUBLE OCCUPANCY

<D> total : 4.000000
<D> / site : 0.800000

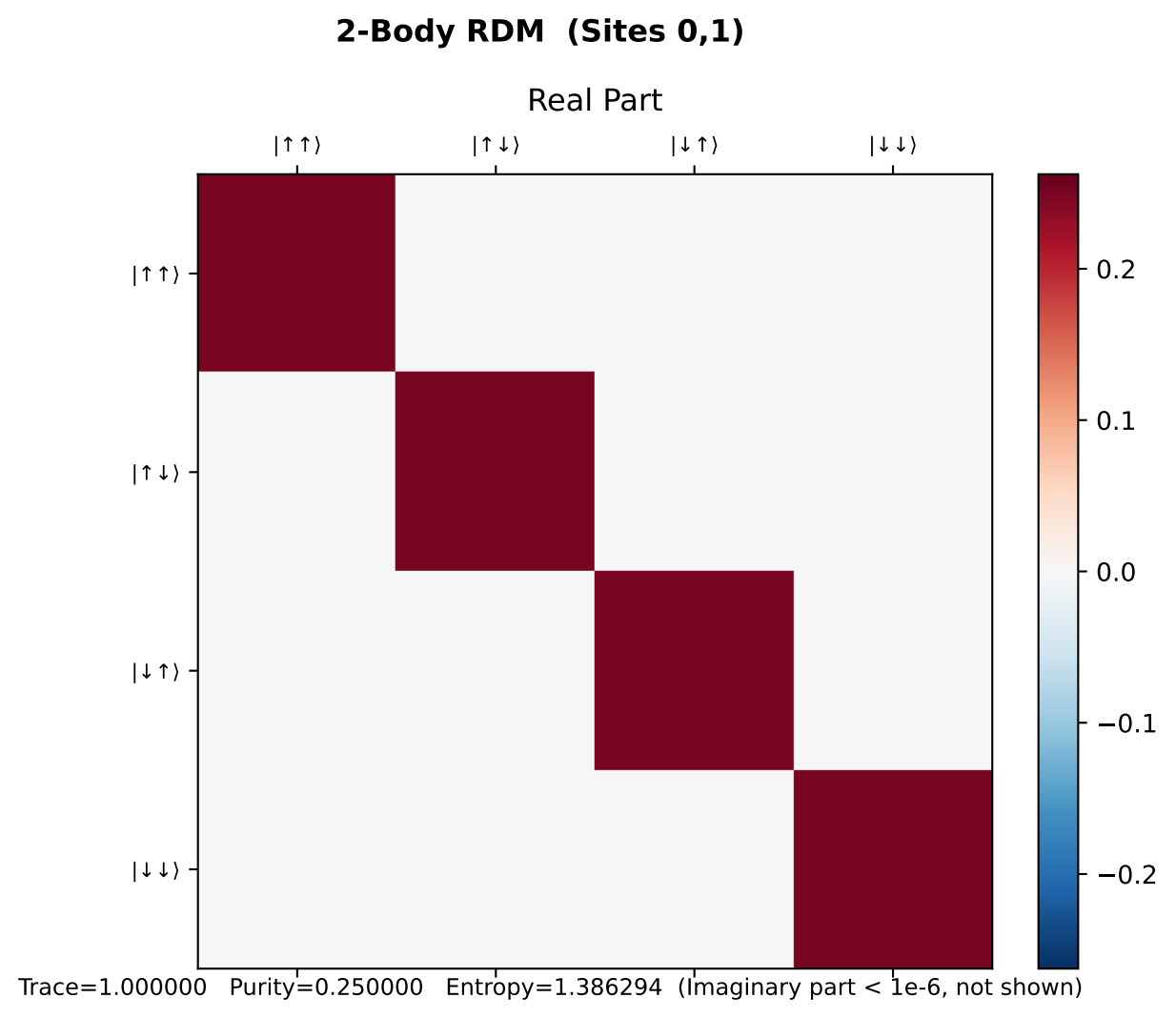
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Body order : 2
Clusters   : 10

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Summary — 2-Body RDMs (Ethylene, CAS(8,5))

Orbitals	Purity	C_GME (pure)	C_LB	C_gcg	C_stiefel	C_UB	State
0,1	0.250000	—	—	—	—	—	MIXED
0,2	0.250000	—	—	—	—	—	MIXED
0,3	0.250000	—	—	—	—	—	MIXED
0,4	0.250000	—	—	—	—	—	MIXED
1,2	0.250000	—	—	—	—	—	MIXED
1,3	0.250000	—	—	—	—	—	MIXED
1,4	0.250000	—	—	—	—	—	MIXED
2,3	0.250000	—	—	—	—	—	MIXED
2,4	0.250000	—	—	—	—	—	MIXED
3,4	0.250000	—	—	—	—	—	MIXED


2-Body RDM (Sites 0,1)
Real Part
|↑↑⟩
|↑↓⟩
|↓↑⟩
|↓↓⟩
0.2
0.1
0.0
−0.1
−0.2
Trace=1.000000 Purity=0.250000 Entropy=1.386294 (Imaginary part < 1e-6, not shown)

Analysis Summary - 2-Body RDM (Sites 0,1)

RDM Metrics
==========

Trace(rho)	: 1.0000000000
Purity Tr(rho^2)	: 0.2500000000
Von Neumann entropy	: 1.3862943611
Eigenvalue min	: 2.500000e-01
Eigenvalue max	: 2.500000e-01
Imaginary part	: < 1e-6 (not shown)

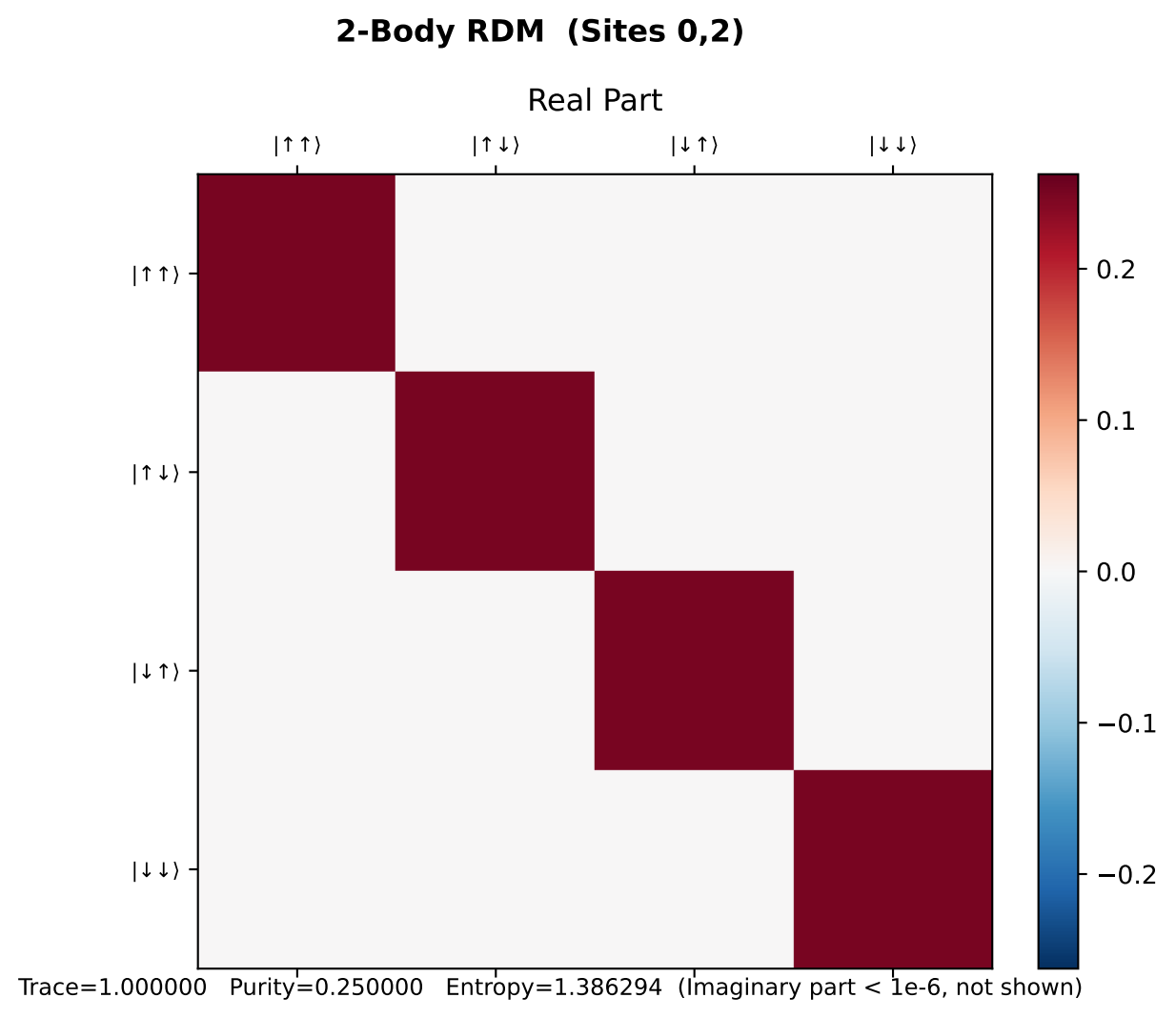
GME Analysis
==========

gamma = Tr(rho^2) = 0.25000000

A	B	Tr(rho_A^2)

STATUS: MIXED STATE

Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.


2-Body RDM (Sites 0,2)
Real Part
|↑↑⟩
|↑↓⟩
|↓↑⟩
|↓↓⟩
0.2
0.1
0.0
−0.1
−0.2
Trace=1.000000 Purity=0.250000 Entropy=1.386294 (Imaginary part < 1e-6, not shown)

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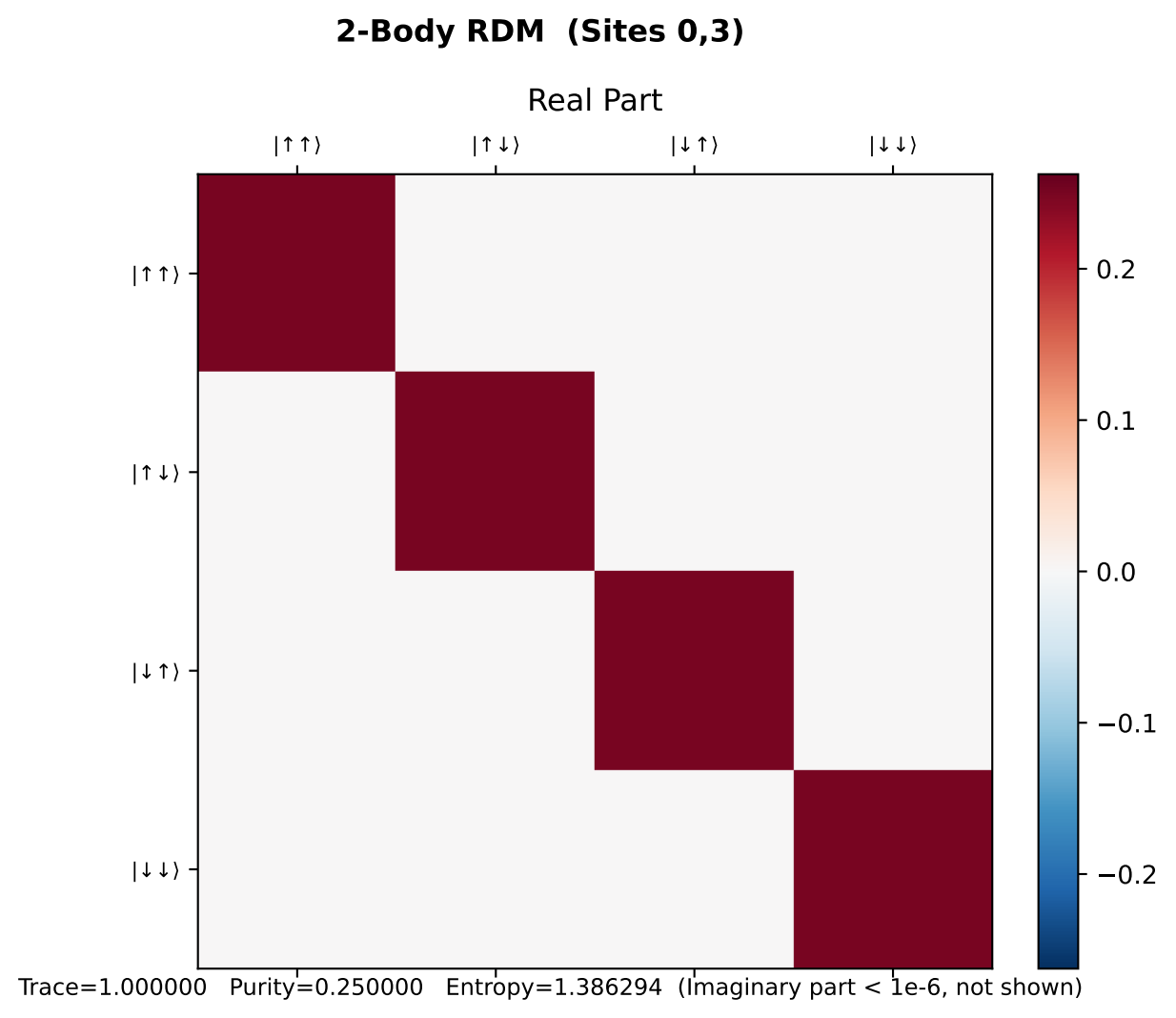
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A	B	Tr(rho_A^2)

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2-Body RDM (Sites 0,3)
Real Part
|↑↑⟩
|↑↓⟩
|↓↑⟩
|↓↓⟩
0.2
0.1
0.0
−0.1
−0.2
Trace=1.000000 Purity=0.250000 Entropy=1.386294 (Imaginary part < 1e-6, not shown)

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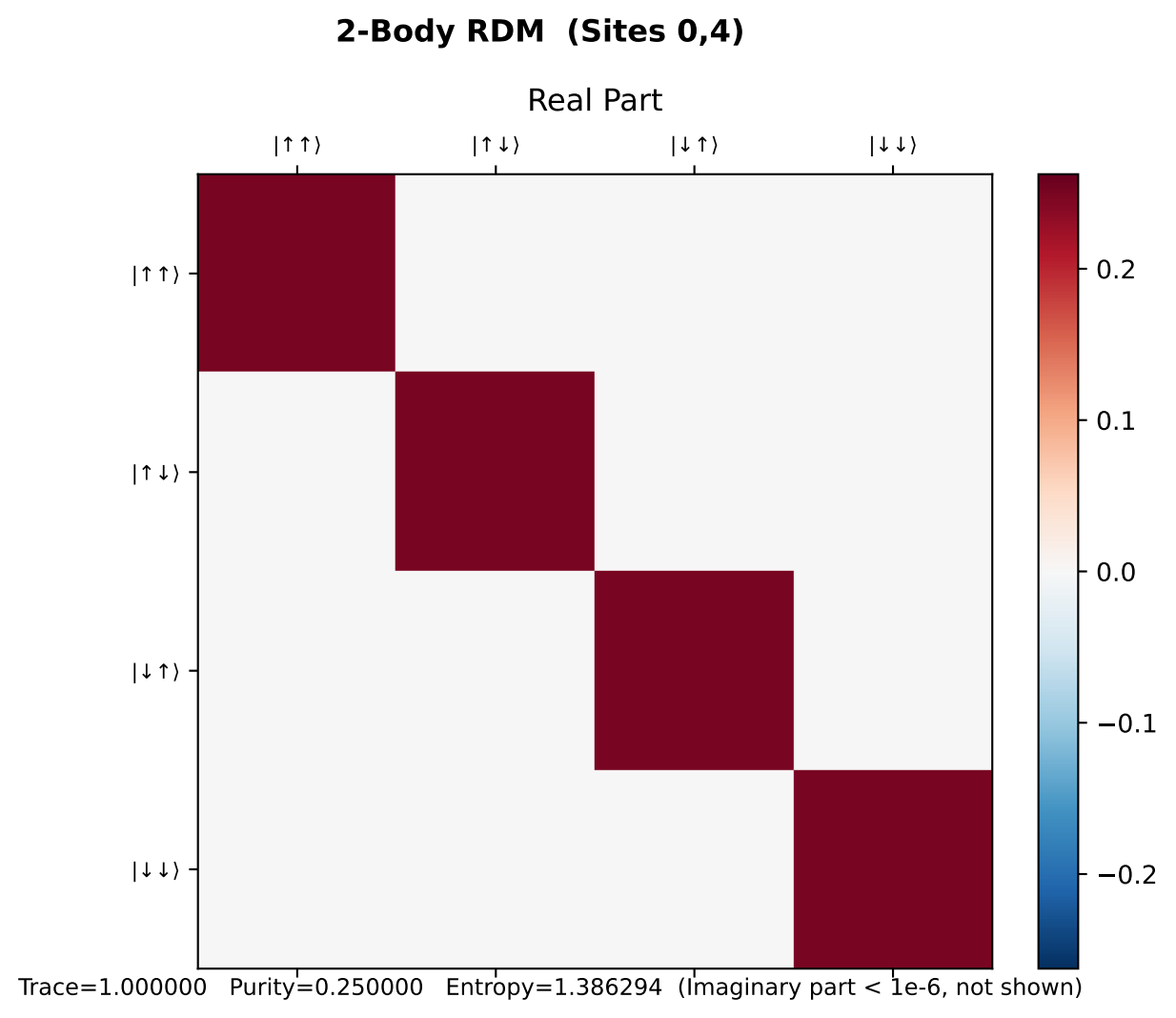
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2-Body RDM (Sites 0,4)
Real Part
|↑↑⟩
|↑↓⟩
|↓↑⟩
|↓↓⟩
0.2
0.1
0.0
−0.1
−0.2
Trace=1.000000 Purity=0.250000 Entropy=1.386294 (Imaginary part < 1e-6, not shown)

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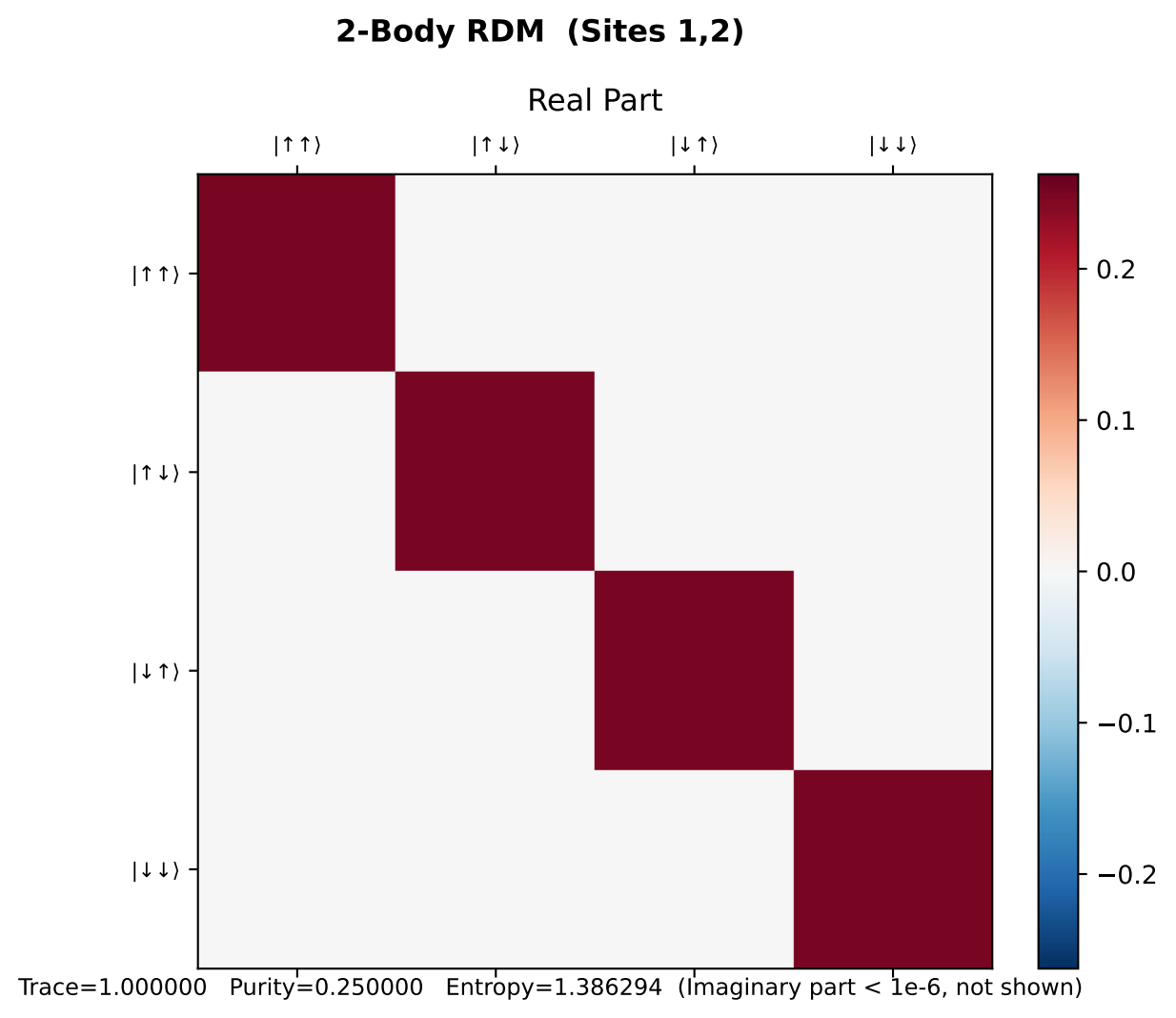
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2-Body RDM (Sites 1,2)
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|↑↑⟩
|↑↓⟩
|↓↑⟩
|↓↓⟩
0.2
0.1
0.0
−0.1
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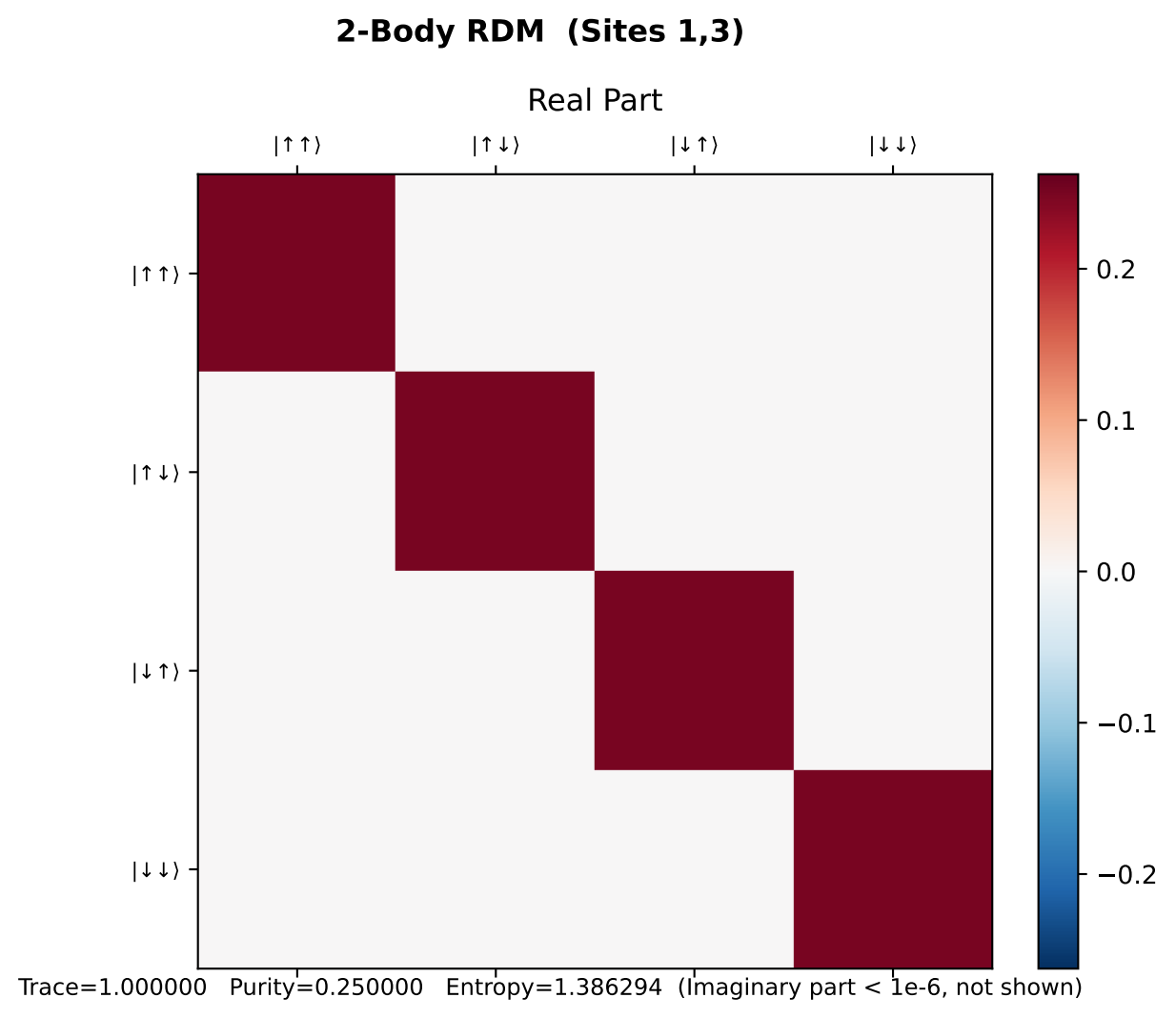
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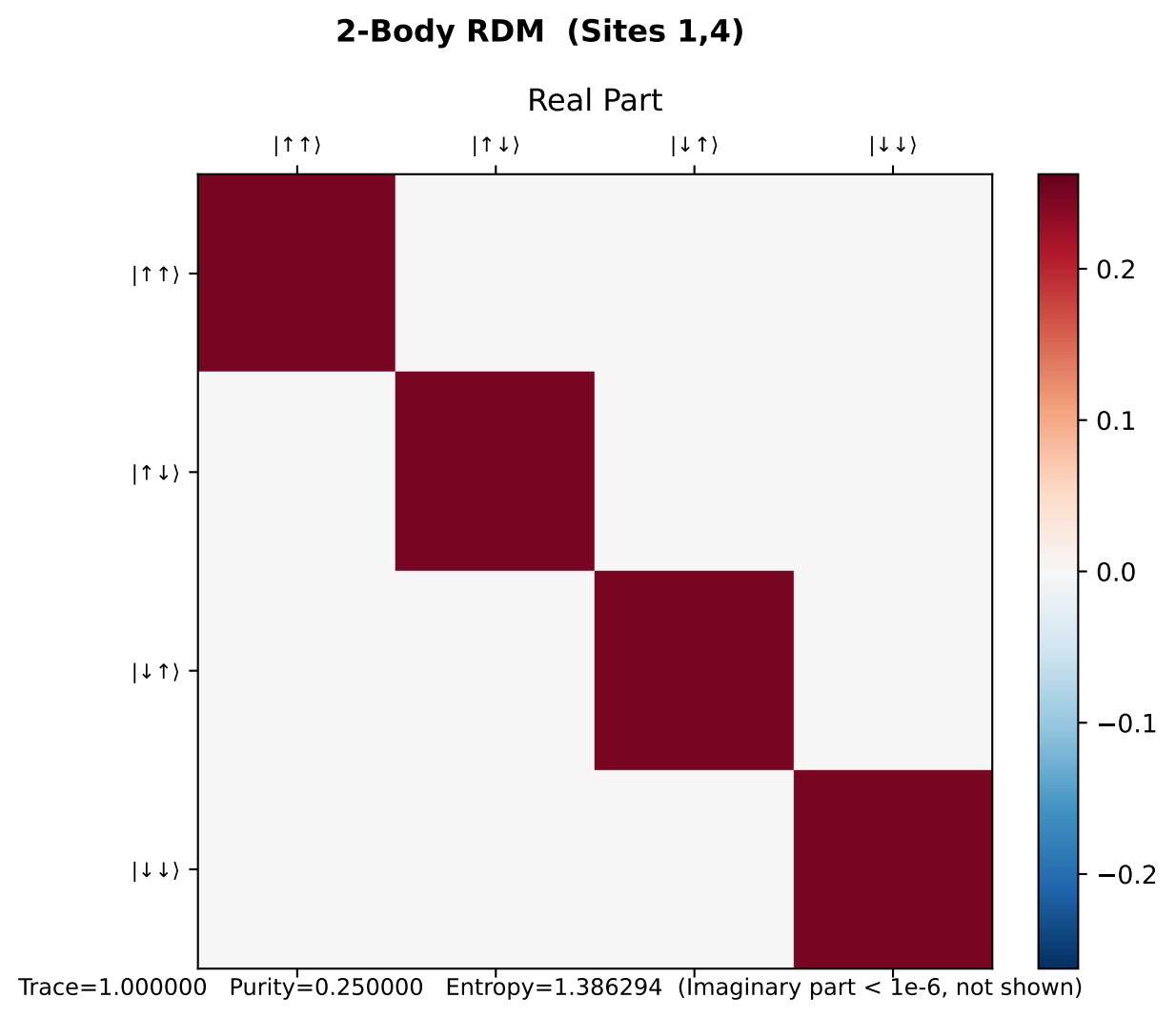
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|↑↓⟩
|↓↑⟩
|↓↓⟩
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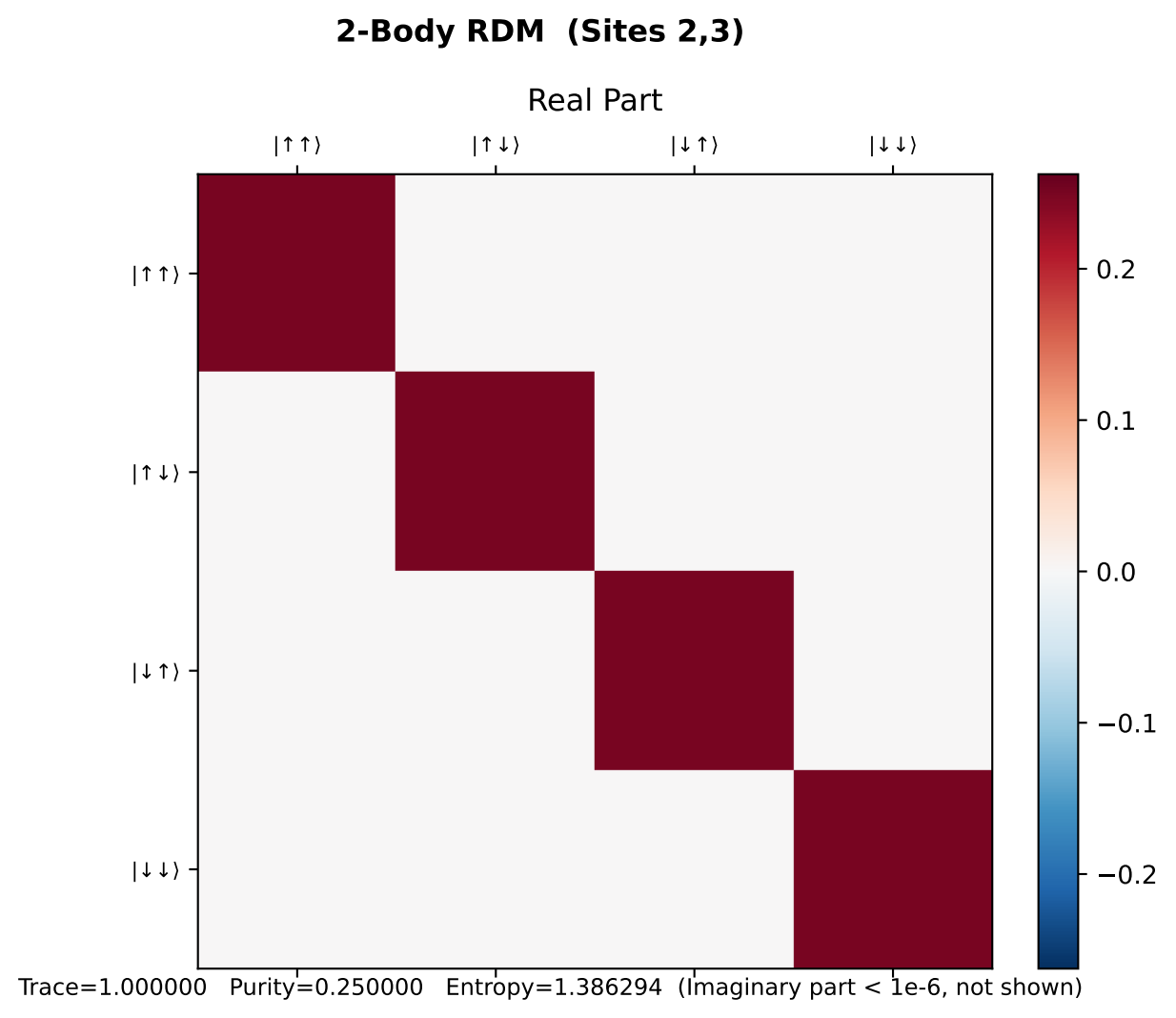
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Real Part
|↑↑⟩
|↑↓⟩
|↓↑⟩
|↓↓⟩
0.2
0.1
0.0
−0.1
−0.2
Trace=1.000000 Purity=0.250000 Entropy=1.386294 (Imaginary part < 1e-6, not shown)

Analysis Summary - 2-Body RDM (Sites 2,3)

RDM Metrics
==========

Trace(rho)	: 1.0000000000
Purity Tr(rho^2)	: 0.2500000000
Von Neumann entropy	: 1.3862943611
Eigenvalue min	: 2.500000e-01
Eigenvalue max	: 2.500000e-01
Imaginary part	: < 1e-6 (not shown)

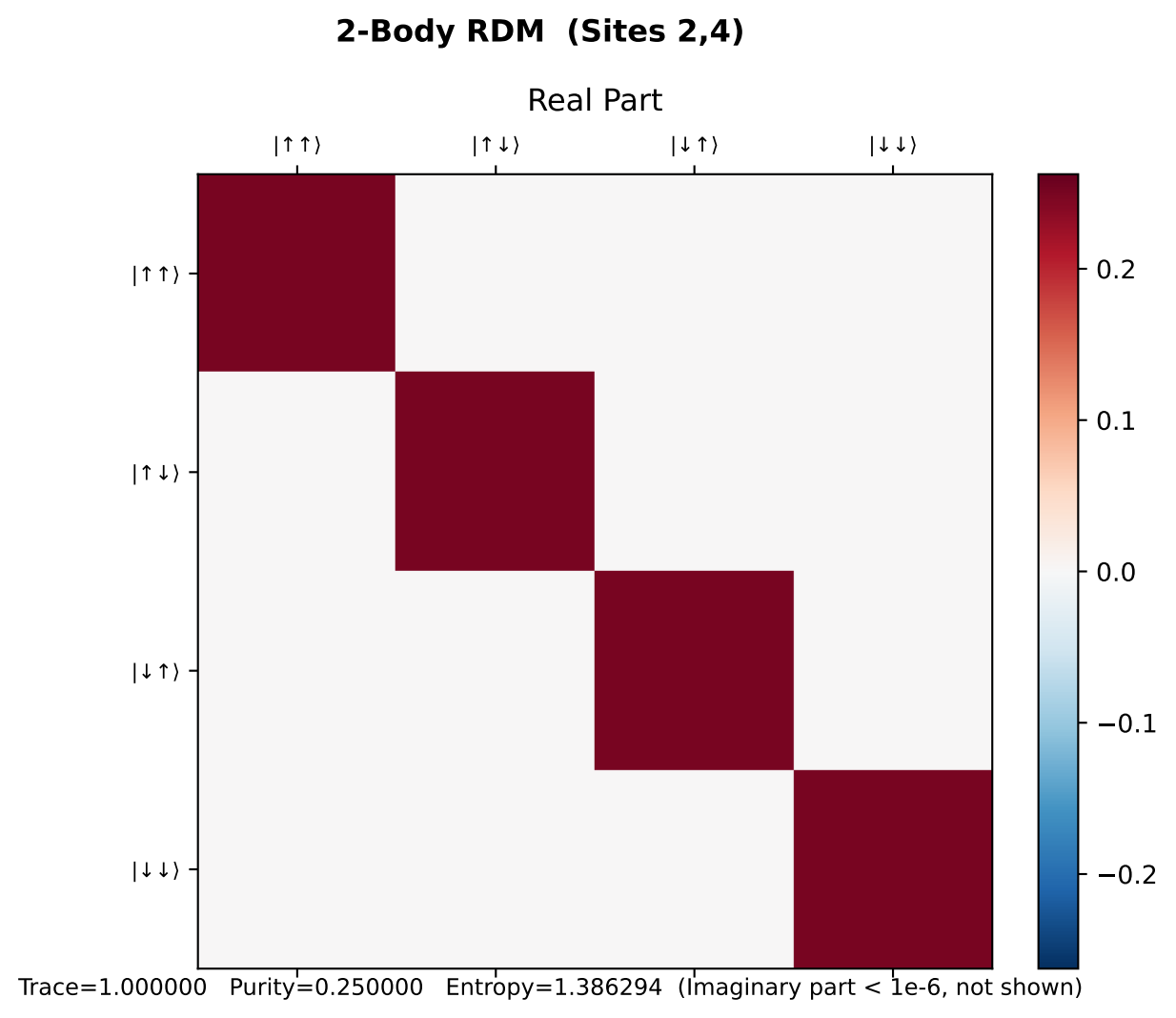
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|↑↓⟩
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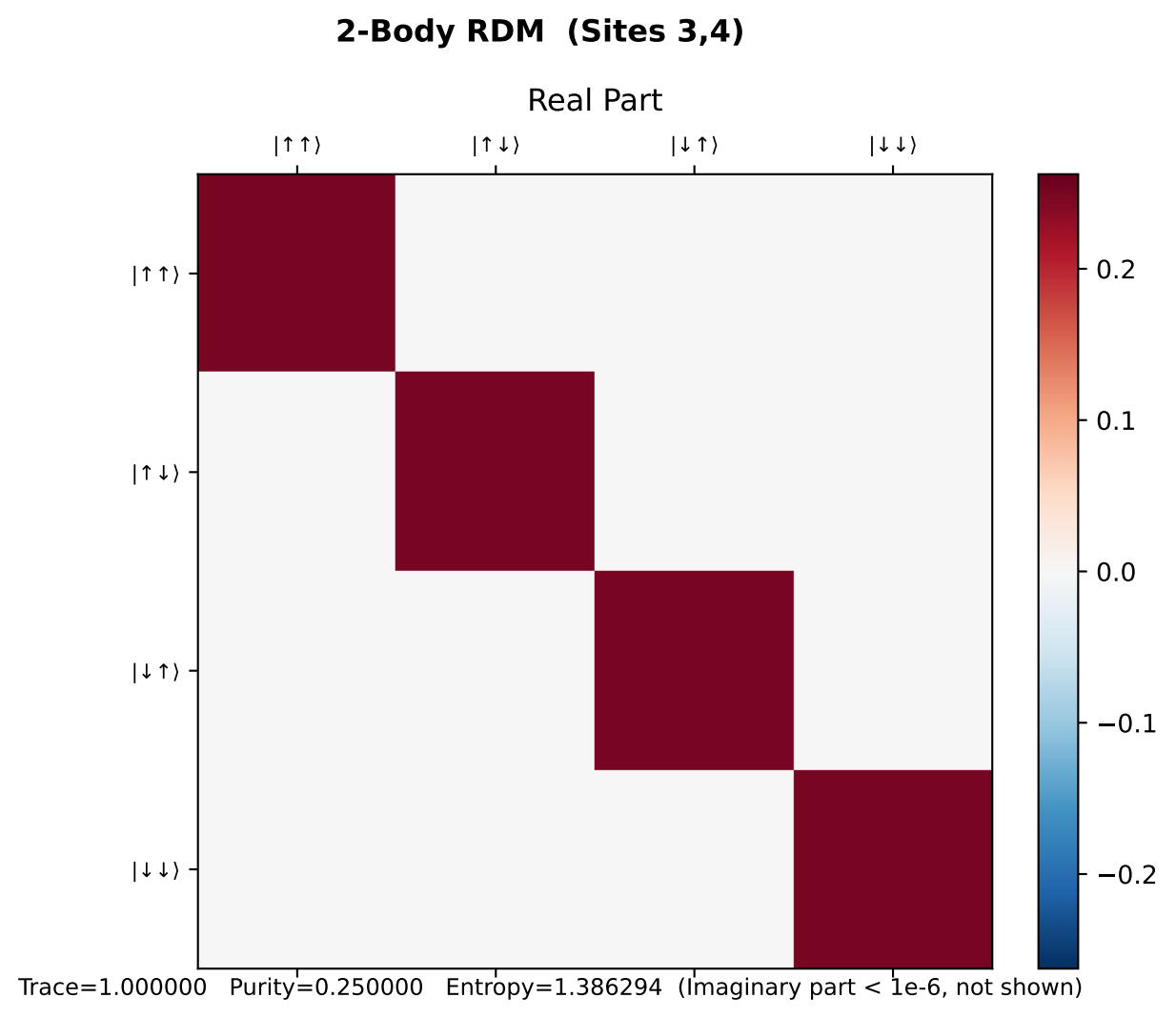
GME Analysis
==========

gamma = Tr(rho^2) = 0.25000000

A	B	Tr(rho_A^2)

STATUS: MIXED STATE

Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.


2-Body RDM (Sites 3,4)
Real Part
|↑↑⟩
|↑↓⟩
|↓↑⟩
|↓↓⟩
0.2
0.1
0.0
−0.1
−0.2
Trace=1.000000 Purity=0.250000 Entropy=1.386294 (Imaginary part < 1e-6, not shown)

Analysis Summary - 2-Body RDM (Sites 3,4)

RDM Metrics
==========

Trace(rho)	: 1.0000000000
Purity Tr(rho^2)	: 0.2500000000
Von Neumann entropy	: 1.3862943611
Eigenvalue min	: 2.500000e-01
Eigenvalue max	: 2.500000e-01
Imaginary part	: < 1e-6 (not shown)

GME Analysis
==========

gamma = Tr(rho^2) = 0.25000000

A	B	Tr(rho_A^2)

STATUS: MIXED STATE

Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.